

JAYDEN SMITH

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SUMMARY

PhD student in Mechanical Engineering specializing in robotics and control systems, with hands-on experience in UAV autonomy, additive manufacturing, and robotic system design. Seeking to apply research expertise to real-world automation and robotics challenges in industry.

TECHNICAL SKILLS

Software: ROS2, OpenCV, MATLAB/Simulink, Python, CasADi, LTspice

CAD: NX, SolidWorks, Autodesk Inventor

Fabrication: Manual and CNC Machining, 3D Printing

EDUCATION

Rensselaer Polytechnic Institute, Troy, NY

Aug 2025 – Present

Doctor of Philosophy in Mechanical Engineering | Concentration: Robotics and Controls

Boeing Fellowship for Rensselaer Graduate Students

Rensselaer Polytechnic Institute, Troy, NY

Aug 2021 – May 2025

Bachelor of Science in Mechanical Engineering | GPA: 3.5/4.0

Minors: Electrical Engineering; Philosophy of Logic, Computability and Mind

Constellation Energy Scholarship | Accelerated BS/PhD Program

WORK EXPERIENCE

The Design Lab, RPI, Troy, NY

May 2025 – Aug 2025

Robotics Intern

- Developed functional modules for both manual and automated control of a Fanuc 200iD LR Mate robot, since used to support capstone students in automation projects
- Designed, manufactured, and tested a suction cup end effector for pick and place operations

RESEARCH EXPERIENCE

Center for Smart Convergent Manufacturing Systems (CSCMS), RPI

Jul 2025 – Present

Research Assistant

- Developing data-driven and physics-based models for adaptive closed-loop control of the Wire-arc Additive Manufacturing (WAAM) process using Computer Vision

Intelligent Systems, Automation and Control (ISAaC) Lab, RPI

Jan 2023 – Present

Research Assistant

- Ran 100 simulations of a Quadrotor Biplane avoiding obstacles and replanning trajectories; data contributed to a peer-reviewed journal paper
- Created an Autoregressive (AR) model of aerial drogue position for use in autonomous aerial refueling

PUBLICATIONS

McIntosh, K., Smith, J., & Mishra, S. (2024). *Aerodynamics-Aware Design and Analysis of Controllers for Tailsitter Vehicles*. IEEE AIM 2024.

PROJECTS

Robotic Art Installation - Project Team Lead

Jan 2023 – Present

- Designing and manufacturing a drawing system of four 3-DOF robot arms with a rotating canvas; developing an MPC to coordinate all four end effectors simultaneously
- Coordinating three sub-teams across mechanical design, controls, and fabrication

CRC-3 Quadrotor Biplane Testbed - Research Assistant

Jan 2023 – May 2025

- Fabricated the 3lb CRC-3 Quadrotor Biplane testbed and performed hardware validation of the onboard model-based path planner and control architecture for autonomous flight

LEADERSHIP & AWARDS

- Graduate Representative & Alumni President, RPI Student Government (Apr 2025 – Present): advocate for graduate student policy and facilitate events averaging 80–150 attendees
- Rensselaer Polytechnic Institute Accelerated BS/PhD Program in Mechanical Engineering (Aug 2024)